“Challenge Loop”
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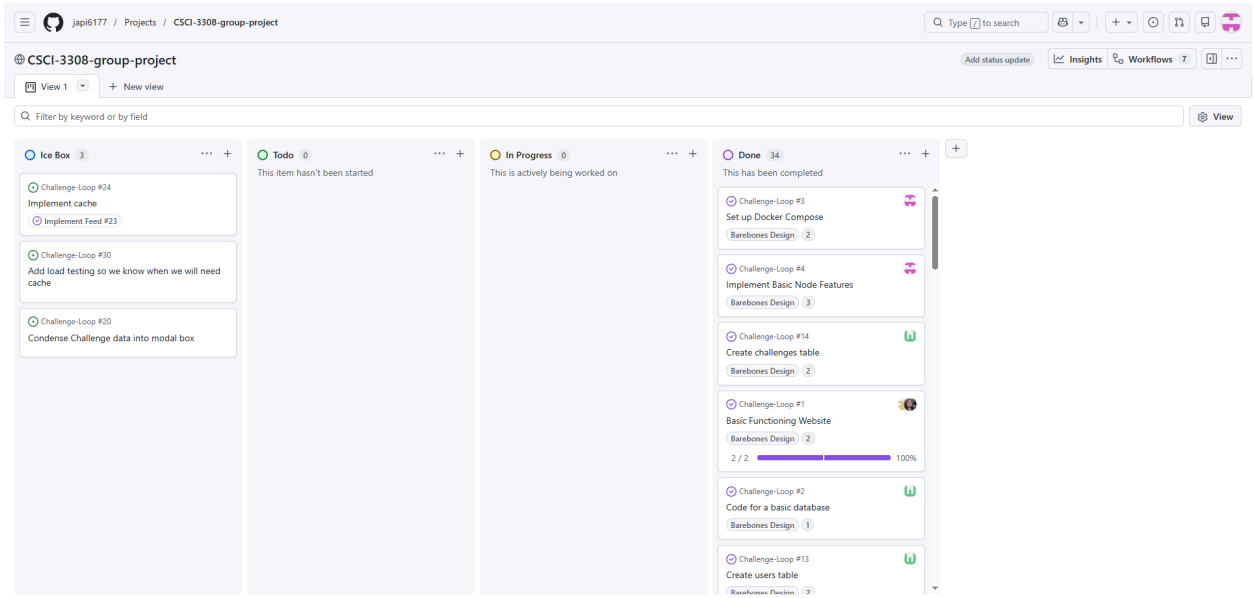
ChallengeLoop is a web application allowing users to join, participate, and compete in different weekly challenges ranging from fitness challenges, productivity challenges, to educational challenges. Users are able to join different challenges in the discover page, record their progress at those challenges, and view their position on the leaderboard. Groups of users are also able to pick and choose their own challenges in a group setting, competing in a friendly setting.

The dashboard summarizes all of the user’s challenges they are actively participating in, along with their overall performance in past challenges which will incentivize them to keep using the app and challenge themselves. Additionally, community engagement through competing is a core focus, allowing users to celebrate their winnings, achievements, and progress they have made. ChallengeLoop uses a simple interface to allow users to stay motivated and not be overwhelmed.

While most habit or goal tracking apps exist for long term habits and consistent challenges, ChallengeLoop provides more realistic challenges that are more short term that still provide users with healthy challenges, while still being fun and competitive. A streak system better helps the user to consistently strive to complete their challenges on a weekly basis. Overall the community promotes engagement between users and motivates the user to a better self.

Project Tracker:
https://github.com/users/japi6177/projects/1/views/1



Video: https://youtu.be/9atXLEkcwWY

Contributions:

Jacob: I implemented social features (add/remove friends, customized social page), UI/UX features (responsive design, visual overhaul), designed the database model, and implemented testing features. I also took on an unofficial directorial role as the project went along, making sure everyone knew what features they were working on, splitting the work-load evenly, and clearing up misunderstandings. Keeping things coordinated and focused ended up being essential to the final result

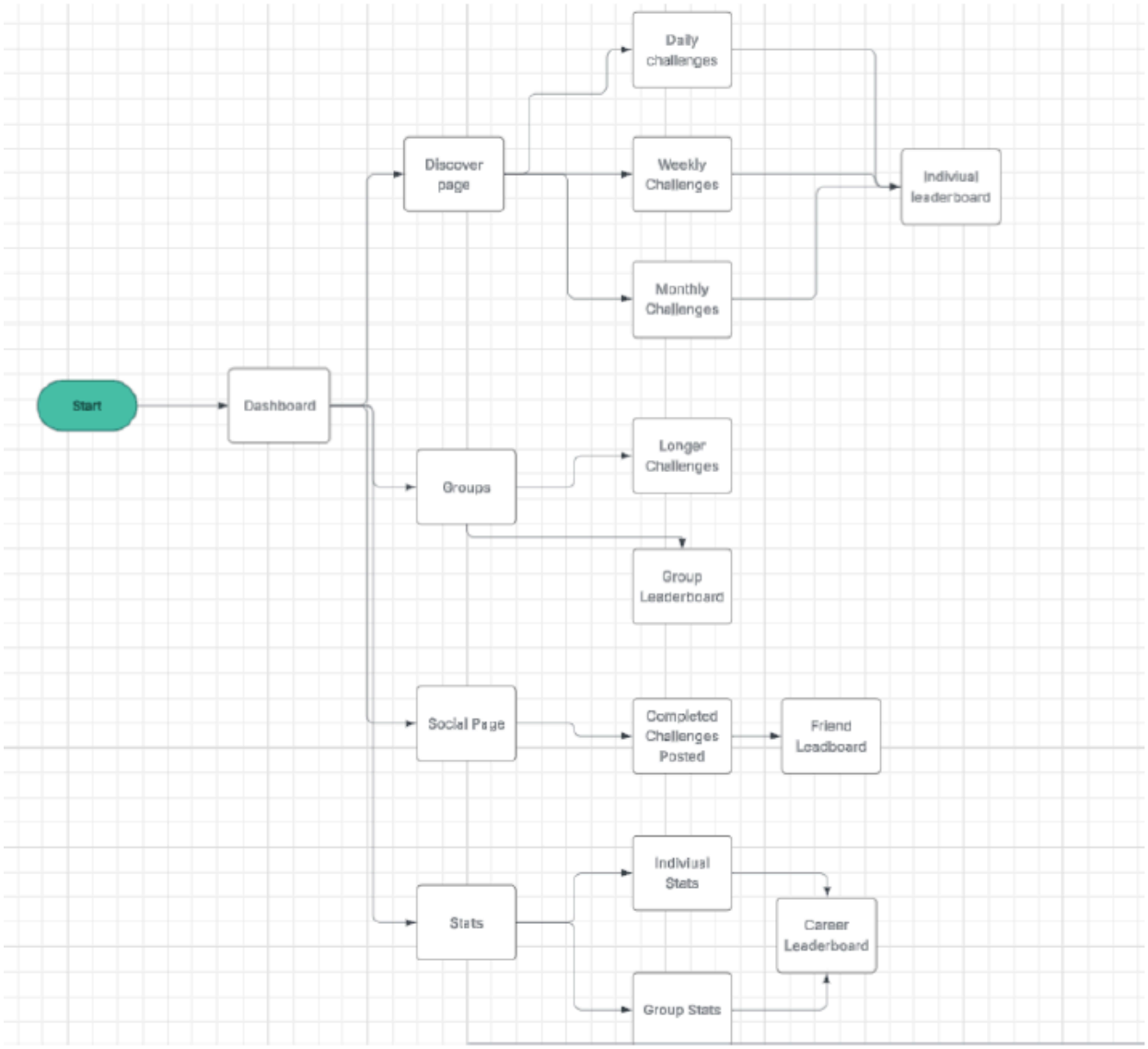
Hunter: Played a role in initial project planning and developing the project's core idea. Provided ideas how the app should function as a system. Operated on backend development with Node.js, Express, PostgreSQL, along with setting up database environments. Implemented frontend and backend key features (posting challenges, commenting functionality, profile page features that display your challenges, and ability to create custom challenges).

Peter: I built the email login system, along with programing the style and theme for the frontend. I also contributed the initial database design. I also setup the render hosting and the email api accounts. Finally I integrated the Gemini API for natural text entry when creating challenges.

Fyn: I built a challenge judging system requiring photo evidence where only approved submissions count toward the leaderboard. I integrated the Groq API to power an AI challenge recommender that analyzes user history. I also added challenge editing and deletion, fixed duration logic, expanded the discover page with 90 pre-built challenges, and implemented profile customization including password changes, email updates, and profile pictures.

Tahnee: I suggested several features, which are currently implemented in the current iteration of our project. I developed our initial application with HTML and Bootstrap. I also played a big part in creating the social page layout, including activity feed, user profile, and a leaderboard for friends. In terms of backend, I created a table for friends which was used to query for statistics for leaderboard purposes using PostgreSQL and I also queried user statistics.

Use-case diagram:



Wireframes: See DatabaseWireframe_3.pdf, ProjectWireframe_3.pdf, and

Test Results: See Lab_10_UAT_Results.pdf

Deployment Link: https://challenge-loop.onrender.com/

Extra Credit:

2. Creativity of the Website from the UI/UX perspective - Jacob

i. Visual Design
The visual design of our app is meant to look like you're in the middle of a forest, looking out through frosted glass. The background is animated to look like light shining through swaying branches. I felt like this would be really relaxing and encourage users to keep coming back, as well as fit in with current design trends like iOS 26. Some icons have been given a slight neon glow, to draw your eye to the most relevant parts of the page.

ii. Navigation
When we were early in development, the navigation was extremely clunky. Adding a challenge sent you to the home-page, you couldn't click comments to navigate to user profiles, the most interesting content was always pushed beneath some irrelevant filler content, etc. I kept a careful record of all of these annoyances as I encountered them and fixed them on the final day of development so I could eliminate as many as possible.

iii. Custom Styling
I think the frosted glass cards and animated background really speak for themselves. There are very few websites that look like this.

iv. Ui Engagement
Almost everything you can click on gives you some kind of feedback, and if not, it was an intentional decision to reduce visual noise. The buttons and cards and pill boxes are all so fun to hover over. They almost look like candy, especially with their neon colors. The most important parts of each page (eg. "Save Entry" on the challenge page) are the ones with the biggest highlights/hover animations, while smaller ones or persistent elements (like most of the navbar) are less dynamic. The website also responds very well to resizing. It's fun to stretch it around and see what it does.

v. Accessibility
I am slightly colorblind so I chose front-UI colors that contrasted well with the background. Title text is usually white with green highlights (if all the text was green it would be hard to read, but this way it stands out just the right amount), less important text is light grey color, and purple is used very sparingly in the leaderboard and the category tags to set them apart from other elements.

3. Responsible AI integration - Fyn and Peter
See "genAI_usage_extra_credit.txt"